GRADUATION PROJECT / SYSTEM ANALYSIS & DESIGN

SmartListing Copilot

Database Design Specification

Multimodal Listing Quality Control for Fashion E-Commerce

DOCUMENT CONTROL	DETAIL
Responsible member	Member 2 — Database Design
Document status	Proposed design for academic/team review
Version	1.1 — 28 October 2026 · seven additional annotated design diagrams
Primary basis	Final Revised FRS (FR-001 to FR-025), 16 October 2026
Architecture basis	Member 3 System Architecture (27 October 2026; pending approval)
Database engine	PostgreSQL — explicitly provisional assumption
Reference categories	Five classes — final labels pending EDA
Validation basis	Schema cross-check and SQLite portability/behavior checks; PostgreSQL execution not demonstrated

DESIGN BOUNDARY

Eight normalized relational tables retain listing inputs, analysis evidence, review findings, controlled suggestions and explicit seller decisions. Images, model weights and lexicons remain outside the relational database. There are no user accounts or marketplace-publishing pathways in the MVP.

Submission integrity

The source FRS is labeled final revised but not stakeholder-approved; the architecture is a team-review proposal. This specification therefore distinguishes sourced requirements from proposed database decisions and leaves the unselected five categories unseeded.

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Reading convention

Confirmed = explicitly stated in supplied FRS/planning. Proposed = database design decision requiring integration review. Open = unresolved elsewhere in the project. Runtime-tested = actual command executed; not inferred from syntax.

01 Executive design overview

The proposed MVP uses an intentionally compact eight-table relational model. An immutable product_listing snapshot references one dataset-sourced image file and records text provenance; each analysis_run is independently repeatable and has a queue state. Separate visual_prediction and text_claim rows retain evidence, and analysis_finding stores bounded warnings. A suggested_correction is optional and can receive at most one seller_decision. A supported_category reference table is intentionally empty until EDA confirms the five names.

DESIGN INDICATOR	RESULT
Physical tables	8
Column definitions	64
Relational foreign keys	7

DESIGN INDICATOR	RESULT
Indexed access patterns	Seven indexes including selected-prediction uniqueness
Core constraints	PK, FK, CHECK, NOT NULL, UNIQUE and guarded defaults
Excluded domains	Accounts, payment, marketplace publishing, price/return/material predictions

02 Scope and source documents

Source hierarchy applied for this package: the FRS is the detailed requirements baseline; revised planning expresses the authoritative scope already referenced by the FRS; Member 3 architecture gives a proposed component/storage boundary. No official System Analysis & Design grading rubric or the original revised-scope proposal was supplied separately.

SOURCE	EVIDENCE STATUS	USE IN THIS DESIGN
Requirements Documentation (DOCX; 16 Oct 2026)	Final revised, stakeholder approval pending	Exact FR-001…FR-025 titles and statements; NFR-SEC/REL/TEC, UC-13 and open items.
Member 1 Project Planning Revised (PDF; 16 Oct 2026)	Scope-aligned planning	Single-image intake, five EDA classes, Product Type and Color; human decisions.
Member 3 Architecture (PDF; 27 Oct 2026)	Design proposal; team review pending	FastAPI/Streamlit, typed module outputs, database versus Model & Lexicon Store split.
Official rubric and original Revised Scope proposal	Not supplied separately	Cannot claim satisfaction or independently resolve rubric/scope conflicts.
Member 1 UML, Member 4 API, Member 5 UI/UX, Member 6 tests	Not supplied in this upload	Integration checks are provisional; require team confirmation.

03 Data requirements analysis

Persistent data is limited to information required to retrieve a submission, reconstruct analytical evidence, present a review result, or document an explicit seller choice. Model inference preprocessing, lexicon contents, rule tables and neural weights are not persisted as relational entities. Text generated for evaluation/demos must be labeled by origin.

Requirements-to-data-needs mapping

REQUIREMENT	PERSISTENCE ENTITY	DATA ELEMENTS	JUSTIFICATION
FR-001 Product Image Upload	product_listing	image_ref, image_origin	Snapshot image reference; actual file-validation process is transient.
FR-002 Product Title and Description Input	product_listing	title, description, text_origin	Original text remains available for review.

REQUIREMENT	PERSISTENCE ENTITY	DATA ELEMENTS	JUSTIFICATION
FR-003 Product Category Input	product_listing	seller_category	Raw category retained even if unsupported.
FR-004 Product Type Classification	visual_prediction; supported_category	attribute, predicted_category_code, model_source, confidence	Primary and CLIP evidence can be compared; weights remain external.
FR-005 Color Recognition	visual_prediction	attribute, predicted_color, confidence, model_source	Color evidence and optional cross-check.
FR-006 Image Quality Analysis	analysis_run; analysis_finding	image_width_px, image_height_px, is_blurred, brightness_status, reason_code	Summaries persisted; raw pixel transforms transient.
FR-007 NLP Claim Extraction	text_claim	attribute, source_field, matched_text, canonical_value	Structured claims; NLP normalization computations transient.
FR-008 English/Arabic Script Detection	text_claim	script_route	Records lexicon routing for extracted claims; per-token detection transient.
FR-009 Image-Text Consistency Checking	analysis_finding; visual_prediction; text_claim	reason_code, visual_value, claimed_value	Stores material differences; comparisons computed by application.
FR-010 Product Type Mismatch Detection	analysis_finding	TYPE_MISMATCH, severity, explanation	Type contradiction evidence.
FR-011 Color Mismatch Detection	analysis_finding	COLOR_MISMATCH, severity, explanation	Color family contradiction evidence.
FR-012 Missing Attribute Detection	analysis_finding	MISSING_ATTRIBUTE, attribute_scope	Optional finding if supporting feature retained.
FR-013 Confidence Scores	visual_prediction; analysis_finding	confidence	Stores available calibrated confidence, not truth guarantee.
FR-014 Low-Confidence Handling	analysis_finding; analysis_run	LOW_CONFIDENCE, queue_state	Threshold configured externally; decisions persisted.
FR-015 Unsupported Category Handling	analysis_finding; analysis_run	UNSUPPORTED_CATEGORY, queue_state	Supported category remains controlled and unknown until EDA.
FR-016 Reason Codes	analysis_finding	reason_code, explanation	Six allowed codes and nonblank explanation.
FR-017 Mismatch Severity Classification	analysis_finding	severity	Rule application is transient; resulting severity persisted.
FR-018 Listing Quality Score	analysis_run	quality_score	Optional 0–100 output; SQL allows NULL if deferred.
FR-019 Listing Inconsistency Level	analysis_run	inconsistency_level	Low/Medium/High result recorded; rule computation external.
FR-020 Review Queue Assignment	analysis_run	queue_state	Queue derived from latest run; no redundant queue table.
FR-021 Template-Based Suggested Fix	suggested_correction	suggested_title, suggested_description	Template output stored; template files remain external.
FR-022 Before/After Comparison	product_listing; suggested_correction	original title/description; suggested_title/description	UI comparison generated from saved snapshots.

REQUIREMENT	PERSISTENCE ENTITY	DATA ELEMENTS	JUSTIFICATION
FR-023 Accept Suggestion	seller_decision	decision=accept, decided_at	Explicit acceptance with no publish capability.
FR-024 Edit Suggestion	seller_decision	decision=edit, final_title, final_description	Stores seller-approved edited wording without changing original.
FR-025 Ignore Suggestion	seller_decision; product_listing	decision=ignore; original title/description	Unapplied suggestion; original input retained.

04 Conceptual data model

Conceptually, a Listing records submitted product content; an Analysis Run produces zero or more visual observations, extracted text claims and findings; optional proposed text is followed by an optional human decision. The supported-category vocabulary describes the five EDA-confirmed product types without assuming names. The Review Queue is a query over the latest relevant completed/partial analysis per listing—not a separate entity.

Visual cross-reference / See Figure 2 for the conceptual cardinalities and the boundary of the eight approved design entities.

Rejected candidates: User (explicitly no authentication); ProductImage (exactly one image per listing); ModelVersion (weights and artifacts reside in the external Model & Lexicon Store). A separate Color table is not justified until the normalized color vocabulary is approved, so color is constrained as a text attribute and controlled by NLP/CV configuration.

05 Entity definitions and relationships

supported_category

Controlled product-type reference list, to be populated only after EDA approves exactly five classes.

Identifier: category_code. Requirements: FR-004, FR-015, FR-017.

Relationships: root lookup; no parent reference.

product_listing

Immutable submitted listing snapshot; one referenced image, submitted text, raw seller category and provenance.

Identifier: listing_id. Requirements: FR-001, FR-002, FR-003, FR-022, FR-025.

Relationships: root lookup; no parent reference.

analysis_run

Repeatable analysis attempt, queue assignment, basic image-quality evidence and reproducibility identifiers.

Identifier: analysis_run_id. Requirements: FR-006, FR-014, FR-015, FR-018, FR-019, FR-020.

Relationships: listing_id → product_listing(listing_id).

visual_prediction

Per-model visual evidence for Product Type or Color; separate baseline/primary/fallback records and one selected record per attribute.

Identifier: prediction_id. Requirements: FR-004, FR-005, FR-013, FR-014, FR-015.

Relationships: analysis_run_id → analysis_run(analysis_run_id); predicted_category_code → supported_category(category_code).

text_claim

Zero or more literal claims extracted from title/description; keeps attribute and origin of each claim.

Identifier: claim_id. Requirements: FR-007, FR-008, FR-009, FR-012.

Relationships: analysis_run_id → analysis_run(analysis_run_id).

analysis_finding

Individual mismatch or triage warning with bounded reason code and optional severity/confidence.

Identifier: finding_id. Requirements: FR-006, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017.

Relationships: analysis_run_id → analysis_run(analysis_run_id).

suggested_correction

At most one controlled proposed title/description for an analysis; optional when unsafe or not generated.

Identifier: suggestion_id. Requirements: FR-021, FR-022.

Relationships: analysis_run_id → analysis_run(analysis_run_id).

seller_decision

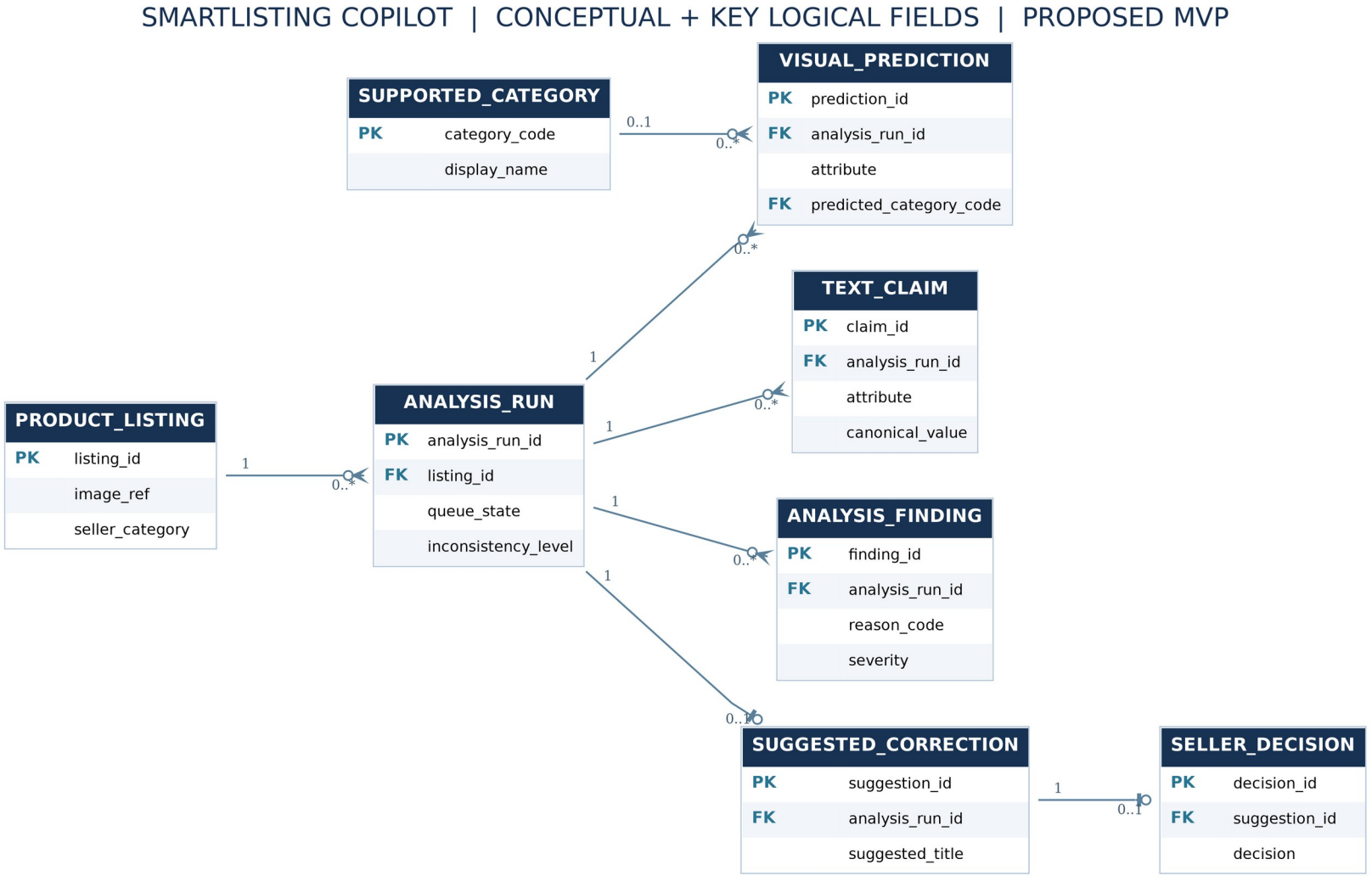
Explicit one-time Accept/Edit/Ignore decision against one proposed correction; retains edited final text only on Edit.

Identifier: decision_id. Requirements: FR-023, FR-024, FR-025.

Relationships: suggestion_id → suggested_correction(suggestion_id).

06 ERD

Figure 1. Crow's-foot relational topology with primary keys (PK), foreign keys (FK) and multiplicities. The full editable schema is supplied in SmartListing_ERD.mmd; physical constraints appear in the SQL files.



07 Logical relational schema

Relations below use [PK] and [FK] labels. Each relation matches the emitted PostgreSQL DDL; non-key columns are omitted from the diagram only for readability, not from the physical schema.

SUPPORTED_CATEGORY (category_code [PK], display_name, is_active)

PRODUCT_LISTING (listing_id [PK], image_ref, image_sha256, image_origin, dataset_product_id, title, description, seller_category, text_origin, created_at)

ANALYSIS_RUN (analysis_run_id [PK], listing_id [FK], status, queue_state, inconsistency_level, quality_score, image_width_px, image_height_px, is_blurred, brightness_status, cv_model_version, nlp_lexicon_version, ruleset_version, created_at, completed_at)

VISUAL_PREDICTION (prediction_id [PK], analysis_run_id [FK], attribute, model_source, prediction_state, predicted_category_code [FK], predicted_color, confidence, is_selected)

TEXT_CLAIM (claim_id [PK], analysis_run_id [FK], attribute, source_field, matched_text, canonical_value, script_route)

ANALYSIS_FINDING (finding_id [PK], analysis_run_id [FK], reason_code, attribute_scope, severity, confidence, visual_value, claimed_value, explanation)

SUGGESTED_CORRECTION (suggestion_id [PK], analysis_run_id [FK], suggested_title, suggested_description, change_explanation)

SELLER_DECISION (decision_id [PK], suggestion_id [FK], decision, final_title, final_description, decided_at)

08 Detailed data dictionary

The authoritative dictionary is the supplied XLSX workbook (64 schema-derived rows with type, precision, nullability, key status, default, constraints, description and example). The compact table below repeats the complete column names and SQL types to support printed reading.

supported_category

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
category_code	VARCHAR(64)	PK / NOT NULL	Stable approved article-type code.
display_name	VARCHAR(120)	NOT NULL	Human-readable class name.
is_active	BOOLEAN	NOT NULL	Whether class can be selected by the model; exactly five active after approval.

product_listing

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
listing_id	BIGINT IDENTITY	PK / NOT NULL	Listing identity.
image_ref	TEXT	NOT NULL	Local or managed-file reference; not image bytes.
image_sha256	CHAR(64)	NULL	Optional checksum for integrity, not a uniqueness constraint.

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
image_origin	VARCHAR(40)	NOT NULL	Sole MVP image dataset provenance.
dataset_product_id	VARCHAR(80)	NULL	Dataset product identifier when available, especially for evaluation splits.
title	TEXT	NOT NULL	Submitted title, kept unchanged for Before/After and audit.
description	TEXT	NOT NULL	Submitted description; empty policy remains pending IN-02.
seller_category	VARCHAR(120)	NOT NULL	Raw seller-supplied category, including possibly unsupported values.
text_origin	VARCHAR(32)	NOT NULL	Submitted text origin for honest dataset/demo provenance.
created_at	TIMESTAMPTZ	NOT NULL	Submission timestamp.

analysis_run

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
analysis_run_id	BIGINT IDENTITY	PK / NOT NULL	Unique analysis attempt.
listing_id	BIGINT	FK / NOT NULL	Analyzed immutable listing.
status	VARCHAR(16)	NOT NULL	Lifecycle status of CV/NLP run.
queue_state	VARCHAR(24)	NULL	Review Queue assignment, set when completed/partial.
inconsistency_level	VARCHAR(8)	NULL	Low/Medium/High heuristic computed from findings.
quality_score	NUMERIC(5,2)	NULL	Optional score 0–100; absent if feature deferred.
image_width_px	INTEGER	NULL	Observed image width, if decoded.
image_height_px	INTEGER	NULL	Observed image height, if decoded.
is_blurred	BOOLEAN	NULL	Blur check result; NULL means not obtained.
brightness_status	VARCHAR(12)	NULL	Observed brightness classification if available.
cv_model_version	VARCHAR(120)	NULL	Identifier of deployed CV weights (not weights themselves).
nlp_lexicon_version	VARCHAR(120)	NULL	Identifier of rules/lexicon artifact (not lexicon content).
ruleset_version	VARCHAR(120)	NULL	Decision rule configuration identifier.
created_at	TIMESTAMPTZ	NOT NULL	Run creation time.

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
completed_at	TIMESTAMPTZ	NULL	Run end time, NULL while running.

visual_prediction

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
prediction_id	BIGINT IDENTITY	PK / NOT NULL	Evidence record identity.
analysis_run_id	BIGINT	FK / NOT NULL	Owning run.
attribute	VARCHAR(16)	NOT NULL	Only allowed visual attributes.
model_source	VARCHAR(20)	NOT NULL	CV, CLIP or dominant-color corroboration.
prediction_state	VARCHAR(16)	NOT NULL	Predicted or abstained result.
predicted_category_code	VARCHAR(64)	FK / NULL	Approved product type; NULL for color/abstention.
predicted_color	VARCHAR(80)	NULL	Normalized color label; NULL for type/abstention.
confidence	NUMERIC(5,4)	NULL	Confidence 0–1; may be absent on abstention.
is_selected	BOOLEAN	NOT NULL	Chosen visual evidence for a given attribute.

text_claim

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
claim_id	BIGINT IDENTITY	PK / NOT NULL	Extracted claim identity.
analysis_run_id	BIGINT	FK / NOT NULL	Owning run.
attribute	VARCHAR(16)	NOT NULL	Product type or color.
source_field	VARCHAR(16)	NOT NULL	Title or description.
matched_text	TEXT	NOT NULL	Exact source phrase supporting claim.
canonical_value	VARCHAR(120)	NOT NULL	Normalized type/color text (may be unsupported).
script_route	VARCHAR(8)	NOT NULL	English rules or fixed Arabic lexicon.

analysis_finding

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
finding_id	BIGINT IDENTITY	PK / NOT NULL	Finding identity.
analysis_run_id	BIGINT	FK / NOT NULL	Owning run.

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
reason_code	VARCHAR(32)	NOT NULL	One of six approved reason codes.
attribute_scope	VARCHAR(16)	NOT NULL	Product Type, Color, Image or whole Listing.
severity	VARCHAR(8)	NULL	Critical/Major/Minor only for true mismatches; NULL for review flags.
confidence	NUMERIC(5,4)	NULL	Warning confidence when available.
visual_value	VARCHAR(120)	NULL	Observed visual evidence for explanation.
claimed_value	VARCHAR(120)	NULL	Observed text claim for explanation.
explanation	TEXT	NOT NULL	Plain-language grounded explanation.

suggested_correction

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
suggestion_id	BIGINT IDENTITY	PK / NOT NULL	Suggestion identity.
analysis_run_id	BIGINT	FK / NOT NULL	Run that generated the proposal.
suggested_title	TEXT	NOT NULL	Template-generated title, never applied automatically.
suggested_description	TEXT	NOT NULL	Template-generated description, never applied automatically.
change_explanation	TEXT	NOT NULL	Why the proposal differs from the submitted text.

seller_decision

COLUMN	SQL TYPE	KEY / NULL	DESCRIPTION
decision_id	BIGINT IDENTITY	PK / NOT NULL	Decision identity.
suggestion_id	BIGINT	FK / NOT NULL	Target proposal; one decision per proposal.
decision	VARCHAR(8)	NOT NULL	Explicit human choice.
final_title	TEXT	NULL	Seller-approved wording if and only if decision is Edit.
final_description	TEXT	NULL	Full final description in Edit case.
decided_at	TIMESTAMPTZ	NOT NULL	Decision timestamp.

09 Normalization analysis

Normalization is demonstrated using concrete functional dependencies, not just primary-key presence.

Visual cross-reference / See Figure 7 for the specific 1NF → 2NF → 3NF decomposition embodied in the relational design.

Unnormalized form and 1NF

A combined spreadsheet-shaped intake record containing repeating fields such as claim_1/claim_2, image_prediction_1/image_prediction_2, and finding_1/finding_2 would violate 1NF and produce variable-width records. The final model stores each claim, model observation and finding as a distinct atomic row, with one value per cell. File references are scalar paths, not embedded image lists.

2NF and candidate keys

All relations have single-column surrogate PKs except supported_category, whose natural PK is category_code. Consequently no non-key attribute can depend on part of a composite chosen primary key. Alternate candidate keys are (analysis_run_id, attribute, model_source) for visual_prediction, (analysis_run_id, reason_code, attribute_scope) for analysis_finding, analysis_run_id for suggested_correction, and suggestion_id for seller_decision. Their attributes describe the whole alternate key, not an isolated subset of it.

3NF and functional dependencies

RELATION	PRINCIPAL NONTRIVIAL DEPENDENCY
product_listing	listing_id → image_ref, title, description, seller_category, text_origin, created_at
analysis_run	analysis_run_id → listing_id, status, queue_state, inconsistency_level, quality_score, timestamps
supported_category	category_code → display_name, is_active
visual_prediction	prediction_id → run_id, attribute, model_source, predicted values, confidence, selection
text_claim	claim_id → run_id, attribute, source_field, matched_text, canonical_value
analysis_finding	finding_id → run_id, reason_code, attribute_scope, severity, explanation
suggested_correction	suggestion_id → run_id, suggested_title, suggested_description, rationale
seller_decision	decision_id → suggestion_id, decision, final_title, final_description

Transitive dependencies are removed: category display_name is not stored in visual_prediction (it is determined via category_code), listing original text is not copied into analysis_run (run_id → listing_id → text), and suggested text is not copied into seller_decision (decision_id → suggestion_id → suggested text). Selected queue status and inconsistency level are intentionally stored as output of an analysis run; although calculable, they are persisted as immutable-at-completion decision outputs for historical review, not duplicated within the same row under another determinant.

Tradeoff: the Review Queue view derives its latest-run selection at query time to avoid a second stored mutable state. Model-vocabulary and color-family rules stay in versioned configuration files outside the relational schema. Constraint logic alone cannot validate the semantic correctness of a model prediction.

10 Physical database design

PostgreSQL is a provisional relational engine because it supports CHECK constraints, foreign keys, partial unique indexes and straightforward local FastAPI integration. Member 3 explicitly leaves the database engine open. SQL scripts target a newly initialized, empty PostgreSQL database and must be executed in numerical order; existing-schema migrations would require a distinct migration mechanism.

SCRIPT	PURPOSE	RERUN BEHAVIOR
01_create_schema.sql	Eight tables, primary/foreign keys, unique and CHECK constraints	CREATE TABLE IF NOT EXISTS; does not update a modified existing table.
02_constraints_indexes.sql	Seven supporting/unique indexes	CREATE INDEX IF NOT EXISTS; no destructive mutations.
03_seed_reference_data.sql	Explains missing EDA labels without fabricating category rows	Read-only status SELECT; approval-dependent seed deferred.
04_validation_queries.sql	Read-only structural/data diagnostics and latest-run queue example	Safe to rerun; results require actual test records.

11 SQL constraints and indexes

uq_visual_selected_attribute — Prevents two selected visual results for the same attribute/run.

idx_listing_created — Submission timeline.

idx_run_listing_latest — Latest-run query and history.

idx_run_queue — Queue filtering.

idx_finding_run_severity — Warnings by run.

idx_claim_run_attribute — Claim extraction detail.

idx_prediction_run_attribute — Visual evidence detail.

High-value constraints include confidence ranges [0, 1], score [0, 100], a six-code warning vocabulary, a three-state queue vocabulary, attribute-specific mismatch severity, one selected prediction per attribute/run, one proposed correction per run, and one decision per proposal. Empty-title/description rules IN-01/IN-02 remain deliberately unenforced until approved.

12 Data integrity and validation rules

Integrity is shared deliberately: PostgreSQL protects referential correctness and data shape; FastAPI/Pydantic validates media formats, allowable input lengths, reason-code routing and complete analysis payloads; the Scoring & Triage Module owns business logic for Q-01/Q-02. No database CHECK constraint is claimed to prove a computer-vision result correct.

Visual cross-reference / See Figure 6 for the exact cascade/restrict/delete graph, including the protected seller-decision dependency.

CONTROL	LOCATION	DESIGN RULE
Single readable image	FastAPI file intake	One image reference per submitted listing; reject unreadable/missing media before analysis.

CONTROL	LOCATION	DESIGN RULE
Allowed categories	supported_category + CV config	Five named EDA classes must be reviewed and seeded; seller input retained raw for unsupported checks.
Prediction confidence	Database CHECK	0 ≤ confidence ≤ 1 where present; uncertain/unavailable may have NULL values.
Multiple model outputs	UNIQUE run+attr+model	Retain primary CV and CLIP baseline independently.
Finding duplication	UNIQUE run+code+scope	At most one warning per reason code and attribute.
Review triage	Backend rule engine	Q-01 and Q-02 unresolved; DB stores chosen state only.
Seller review	seller_decision	Accept/Edit/Ignore explicit; Edit keeps final approved words; no publishing action.

13 Data lifecycle and deletion behavior

Proposed lifecycle: create an immutable product_listing snapshot → create an analysis_run → append visual predictions, extracted text claims and warnings → set final queue state/score → optionally save a suggestion → optionally save one seller decision. Reanalysis creates a new run, never overwriting prior evidence. Backend writes stages transactionally where appropriate; partial NLP/CV failure must result in Needs review if a partial result is saved.

Visual cross-reference / Figure 4 follows the persistent evidence generated by each run; Figure 5 isolates the mandatory human decision gate.

Deletion is intentionally conservative: deleting a listing with analysis runs is restricted; deleting a run cascades its predictions, claims, findings and suggestion but is blocked if a seller decision references that suggestion; deleting an approved suggestion is restricted. Deletion and retention durations are not approved yet, so no automatic purge job is included. After a retention policy is agreed, administrative purge must use explicit authorization and backup procedures; user accounts are not introduced here.

14 Security and privacy considerations

The prototype is local only, with no external upload or marketplace publishing connector. Avoid SQL image blobs: the database stores image_ref and optional checksum while local file access permissions guard original files. Do not store personal user accounts or credentials. Escape/validate input with Pydantic and bound SQL parameters; never interpolate seller text into SQL. Protect database credentials in environment variables, not checked-in source. Dataset/synthetic evaluation text is distinguished using text_origin.

Member 3 identifies external Model & Lexicon Store files for weights, thresholds, vocabularies, lexicons, group/family tables and templates. Artifact identifiers in the analysis run are for reproducibility; DB is not an approved model registry. No new access control layer is assumed.

15 Requirements traceability matrix

The following matrix preserves the exact supplied functional requirement statements. Coverage is structural capacity, not verified system implementation. “Application” means the persistence layer alone cannot enforce inference/queue behavior.

ID / EXACT FRS STATEMENT	MAPPED RELATION(S)	KEY ATTRIBUTES	COVERAGE
FR-001 The system shall allow a seller to provide one fashion product image for listing analysis.	product_listing	image_ref, image_origin	Proposed — mapped
FR-002 The system shall accept a product title and a product description as text for attribute-claim extraction.	product_listing	title, description, text_origin	Proposed — mapped
FR-003 The system shall accept the seller-provided product category and make it available to the unsupported-category check.	product_listing	seller_category	Proposed — mapped
FR-004 The system shall predict product type within the five EDA-selected categories using the ResNet/MobileNet main CV model, with CLIP zero-shot maintained as baseline and fallback.	visual_prediction; supported_category	attribute, predicted_category_code, model_source, confidence	Proposed — mapped
FR-005 The system shall predict product color from the image, with a dominant-color check used to cross-check the visual color decision.	visual_prediction	attribute, predicted_color, confidence, model_source	Proposed — mapped
FR-006 The system shall perform basic checks of image resolution, blur and brightness and make relevant quality findings available to review and, when the optional Listing Quality Score is enabled, to scoring.	analysis_run; analysis_finding	image_width_px, image_height_px, is_blurred, brightness_status, reason_code	Proposed — mapped
FR-007 The system shall normalize title and description text and extract only product-type and color claims using a controlled vocabulary, rules and fuzzy matching.	text_claim	attribute, source_field, matched_text, canonical_value	Proposed — mapped
FR-008 The system shall identify Arabic characters to select the small fixed Arabic lexicon and otherwise apply the English vocabulary.	text_claim	script_route	Proposed — mapped
FR-009 The system shall compare image-supported product type and color with claims extracted from title and description using rules and a lightweight mismatch classifier.	analysis_finding; visual_prediction; text_claim	reason_code, visual_value, claimed_value	Proposed — mapped
FR-010 The system shall flag contradictions between visual product type and text claims and distinguish same-group from different-group contradictions.	analysis_finding	TYPE_MISMATCH, severity, explanation	Proposed — mapped

ID / EXACT FRS STATEMENT	MAPPED RELATION(S)	KEY ATTRIBUTES	COVERAGE
FR-011 The system shall apply color-family and near-shade rules to flag different-family color contradictions while allowing minor or ignored near-shade differences.	analysis_finding	COLOR_MISMATCH, severity, explanation	Proposed — mapped
FR-012 The system shall identify when an image-supported product type or color is absent from the submitted text rather than contradicted by it.	analysis_finding	MISSING_ATTRIBUTE, attribute_scope	Proposed — optional feature
FR-013 The system shall expose confidence for visual predictions and mismatch findings, distinguishing confidence from verified correctness.	visual_prediction; analysis_finding	confidence	Proposed — mapped
FR-014 The system shall flag predictions below the configured validation-derived confidence threshold and route affected listings to Needs Review rather than asserting certainty.	analysis_finding; analysis_run	LOW_CONFIDENCE, queue_state	Proposed — mapped
FR-015 The system shall flag suspected out-of-scope images when CV and CLIP disagree, the predicted type contradicts the seller category, or CLIP favors the other-product prompt, avoiding a forced definitive type.	analysis_finding; analysis_run	UNSUPPORTED_CATEGORY, queue_state	Proposed — mapped
FR-016 The system shall associate applicable findings with the proposal-defined reason codes and a plain-language explanation template.	analysis_finding	reason_code, explanation	Proposed — mapped
FR-017 The system shall assign severity using the approved product-group and color-family rule table.	analysis_finding	severity	Proposed — mapped
FR-018 The system shall calculate a transparent 0–100 Listing Quality Score from image quality, attribute completeness and image-text consistency only.	analysis_run	quality_score	Proposed — optional feature
FR-019 The system shall calculate a heuristic Listing Inconsistency Level from the number and severity of Major and Critical mismatches.	analysis_run	inconsistency_level	Proposed — mapped
FR-020 The system shall assign each analyzed listing to the proposal-defined review status based on inconsistency, low confidence, image blur, missing attributes and unsupported-category flags.	analysis_run	queue_state	Proposed — mapped

ID / EXACT FRS STATEMENT	MAPPED RELATION(S)	KEY ATTRIBUTES	COVERAGE
FR-021 The system shall propose a corrected title and description through controlled templates grounded only in supported type and color predictions, without using an LLM.	suggested_correction	suggested_title, suggested_description	Proposed — mapped
FR-022 The system shall present original and proposed listing text side by side and indicate what changed and why.	product_listing; suggested_correction	original title/description; suggested_title/description	Proposed — mapped
FR-023 The system shall allow the seller to accept a presented correction explicitly without automatically publishing the listing.	seller_decision	decision=accept, decided_at	Proposed — mapped
FR-024 The system shall allow the seller to modify the proposed text before approving their chosen wording, without autonomous publication.	seller_decision	decision=edit, final_title, final_description	Proposed — mapped
FR-025 The system shall allow the seller to ignore a suggested correction and retain the original listing text for that decision.	seller_decision; product_listing	decision=ignore; original title/description	Proposed — mapped

No FR is declared Not Applicable because all 25 have either a stored input/output or a reproducibility/review concern. Transient-only substeps (script detection, preprocessing, CV execution and scoring computation) are explicitly identified in Section 3 as application operations rather than separate tables.

16 Architecture and API alignment

Member 3 architecture proposes Streamlit Frontend → FastAPI Backend → CV and NLP in parallel → Mismatch Detection → Scoring & Triage → Suggestion; FastAPI alone reads and writes Database. Conceptual entities align to ListingInput, VisualEvidence, TextClaims, MismatchReport, ListingAssessment, SuggestedFix and SellerDecision objects; the names and endpoint definitions are Member 4 decisions, not approved here.

Visual cross-reference / Figure 3 separates transient CV/NLP processing from database rows and external files; Figure 8 explains why the queue is a derived view.

INTEGRATION REQUEST	PERSISTENCE REPRESENTATION	DEPENDENCY
Submit listing	product_listing (image_ref, title, description, seller_category)	Member 4 to decide multipart upload vs encoded payload; IN-01…IN-04 pending.
Retrieve analysis/result	analysis_run + child evidence + finding + optional suggestion	Member 4 Pydantic contracts and failures.
Record Accept/Edit/Ignore	seller_decision linked to suggested_correction	Exact return payload and edit history policy pending.
Review Queue UC-13	Latest completed/partial analysis_run for each listing, grouped by queue_state	Member 5 queue-view workflow and Q-01/Q-02 approval.
NFR evaluation artifacts	External scripts/model files; reference identifiers on run	Member 6 reproducibility data-card/test plan.

17 Assumptions and open decisions

ID	OPEN ITEM	PROVISIONAL TREATMENT	OWNER
D-01	Database engine	PostgreSQL assumed; Member 3 did not select an engine.	Member 2 + Member 3
D-02	Five product types	No actual EDA names supplied; supported_category is unseeded.	CV / Dataset lead
D-03	Q-01/Q-02 triage	Medium-only and overlapping-flag Review Queue precedence unresolved.	FRS owner / Supervisor
D-04	IN-01 to IN-04	Empty title/description, image decode and category error policies not fixed.	Member 4
D-05	Decision retention	Duration, edit-history detail, and deletion authority not approved.	Product team
D-06	Image storage path	Filesystem or object store and upload transfer format not selected.	Member 3 + Member 4
D-07	Category/color labels	Color aliases/families and near-shade behavior are config decisions post-EDA.	Member 4 / CV
D-08	Team deliverables	Member 1 UML, 4 API, 5 UI/UX, 6 test plan not supplied.	Each member
D-09	Source scope conflict	Seller image upload vs dataset-only image provenance requires allowed demo-input decision.	FRS owner
D-10	Review actor	Architecture mentions Catalog Reviewer queue but no reviewer account/auth exists.	Member 5

18 Design verification results

Static cross-artifact review covers eight tables, 64 columns, seven foreign keys, seven indexes and FR-001…FR-025. See Validation_Report.md for mechanically executed results, the SQLite behavior-simulation scope and outstanding PostgreSQL-only execution dependency. SQL cannot be reported as tested against PostgreSQL in this environment unless that engine is actually run.

Acceptance verification plan: DDL parse/compile on an actual PostgreSQL instance; seed the five EDA-confirmed categories; execute positive and negative inserts inside a rollback transaction; verify duplicate primary keys/selected predictions/findings, foreign key violations, check bounds, optional NULLs, multiple runs and warnings, partial-run routing, and RESTRICT/CASCADE behavior; run read-only validation diagnostics and capture evidence.

19 Conclusion

The specification establishes a restrained persistence layer for the approved multimodal review workflow without adding accounts, market publishing, or speculative commercial fields. Once the team confirms the database engine, five category labels and API/queue decisions, the documented schema can be baselined for implementation. Until then, this is an academically reviewable design proposal, not an approved or production-tested database.

20 Appendices

Appendix A — Canonical enumerations

DOMAIN	ALLOWED VALUES
finding.reason_code	TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE, UNSUPPORTED_CATEGORY
analysis_run.queue_state	ready_to_publish, needs_review, critical_mismatch
analysis_run.inconsistency_level	low, medium, high
analysis_run.status	running, completed, partial, failed
visual_prediction.attribute	product_type, color
seller_decision.decision	accept, edit, ignore
text_origin	seller_input, manual_demo, synthetic_evaluation
supported_category	Five EDA-approved labels — intentionally not predefined

Appendix B — Executable artifacts

SQL/01_create_schema.sql; SQL/02_constraints_indexes.sql; SQL/03_seed_reference_data.sql; SQL/04_validation_queries.sql. Editable ERD: SmartListing_ERD.mmd. Data Dictionary: SmartListing_Data_Dictionary.xlsx. Source-of-truth generated from schema_authority.json during packaging (this internal file is not a deliverable).

21 Controlled source interpretation

The precise wording of requirements is taken from the supplied final revised FRS, not reconstructed from prior drafts. The FRS itself identifies the 12 October Revised Scope proposal as governing; that separately authored original proposal and the official grading rubric are absent. The Member 3 architecture explicitly describes itself as a pending proposal, so its data storage boundary is treated as a design-alignment input, not formal authorization.

22 Design decisions and alternatives

Design alternatives rejected: a User table introduces an unapproved authentication requirement; a ProductImage child table suggests multi-image listings; embedding weights/lexicons in PostgreSQL would contradict Member 3 storage separation; a materialized Review Queue risks stale status; a separate ModelVersion registry would be premature. A category lookup is retained because it is the least speculative way to enable the five-class restriction after EDA without inventing class codes.

23 Integration handoff checklist

- Member 1: confirm use cases can reanalyze a listing without mutating its original snapshot.
- Member 3: approve PostgreSQL and local image reference storage.
- Member 4: finalize Pydantic request/response fields, error policy, queue precedence and one-suggestion decision API.
- Member 5: verify original/suggested/edited text and latest Review Queue state can be rendered with this schema.
- Member 6: run actual PostgreSQL negative tests and resolve open issues before claiming end-to-end acceptance.
- Dataset/CV owners: approve exactly five EDA classes and align model output codes before seeding.

End of core specification • version 1.1 • visual architecture atlas follows

24 Visual Architecture Atlas

SEVEN SCHEMA-GROUNDED TECHNICAL PLATES

A visual companion to the authoritative ERD (Figure 1)

The plates below separate database structure, processing provenance, human decisions, and data-governance constraints. They all refer to the same eight-table SQL schema; diagrams do not authorize new entities, features, or product categories.

FIGURE	TECHNICAL VIEW	PRIMARY PURPOSE
02	Conceptual domain model	The eight entities and their business relationships without physical SQL detail.
03	Persistence boundary and component interaction	Which operations are transient and which outputs are stored.
04	Analysis evidence and repeatability	How each listing generates separately retained analysis evidence.
05	Human-controlled correction gate	The one-way boundary between suggested wording and an explicit decision.
06	Referential integrity and deletion safeguards	Parent-child actions represented by the actual SQL foreign keys.
07	Normalization pathway to Third Normal Form	From a repeating-group working record to the final eight relations.
08	Derived Review Queue read model	How the reviewer sees the latest relevant status without a ninth table.

DIAGRAM RULES

Solid connectors express stored relational links or the ordered workflow stated beside them. Dashed connectors represent a processing dependency, optional lookup, or a reference to external artifacts; they are never to be read as additional foreign keys.

Authoritative implementation: SQL/01_create_schema.sql and SQL/02_constraints_indexes.sql. Visual sources: Diagrams/*.dot and Diagrams/*.svg. Figures 2–8 supplement rather than supersede the full ERD in Figure 1.

DESIGN PLATE / FIGURE 02 OF 08

Conceptual domain model

The eight entities and their business relationships without physical SQL detail.

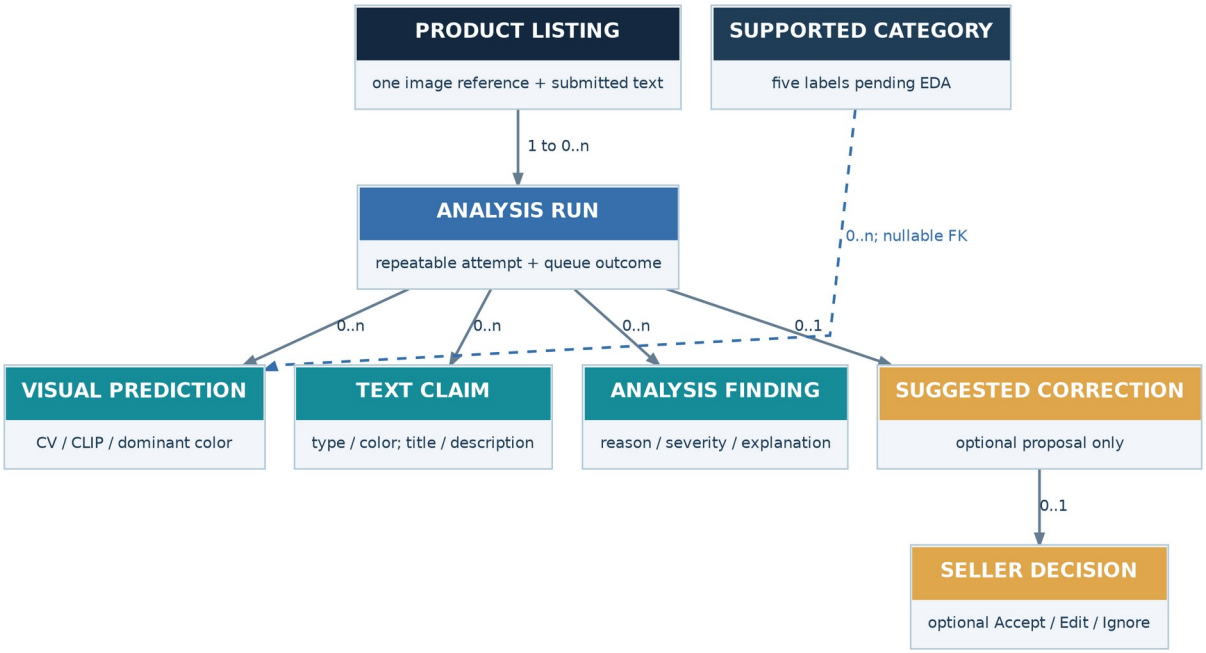


Figure 2. Conceptual domain model.

ENGINEERING INTERPRETATION

The analysis run is the central repeatable event. A seller decision is optional until an explicit seller choice is recorded; categories remain an EDA-dependent controlled vocabulary.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: product_listing; analysis_run; visual_prediction; text_claim; analysis_finding; suggested_correction; seller_decision; supported_category.

Section mapping: 04 Conceptual data model · FR-001 to FR-025

DESIGN PLATE / FIGURE 03 OF 08

Persistence boundary and component interaction

Which operations are transient and which outputs are stored.

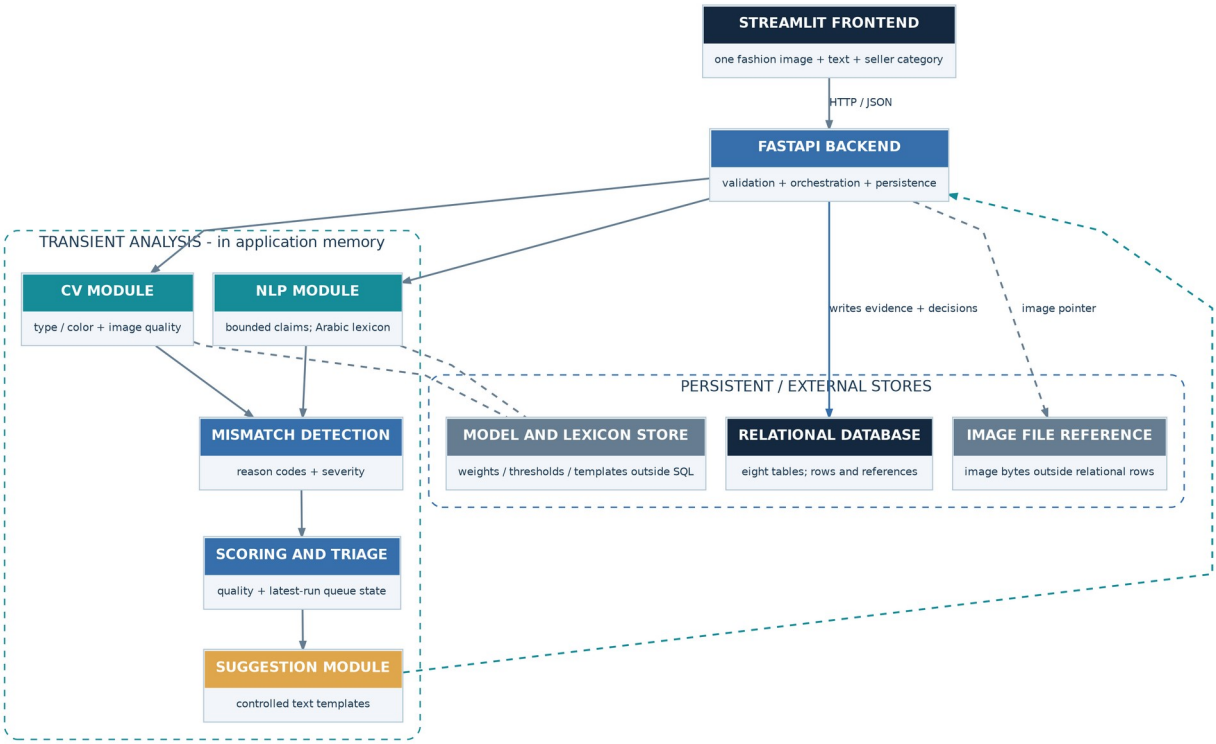


Figure 3. Persistence boundary and component interaction.

ENGINEERING INTERPRETATION

Only the FastAPI Backend is responsible for database persistence. Images, model weights, controlled vocabularies and templates remain external to relational tables. The diagram depicts Member 3's proposed architecture rather than deployed components.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: Member 3 canonical component names; SQL/01_create_schema.sql; source document architecture proposal.

Section mapping: 16 Architecture and API alignment · Member 3 proposed architecture

DESIGN PLATE / FIGURE 04 OF 08

Analysis evidence and repeatability

How each listing generates separately retained analysis evidence.

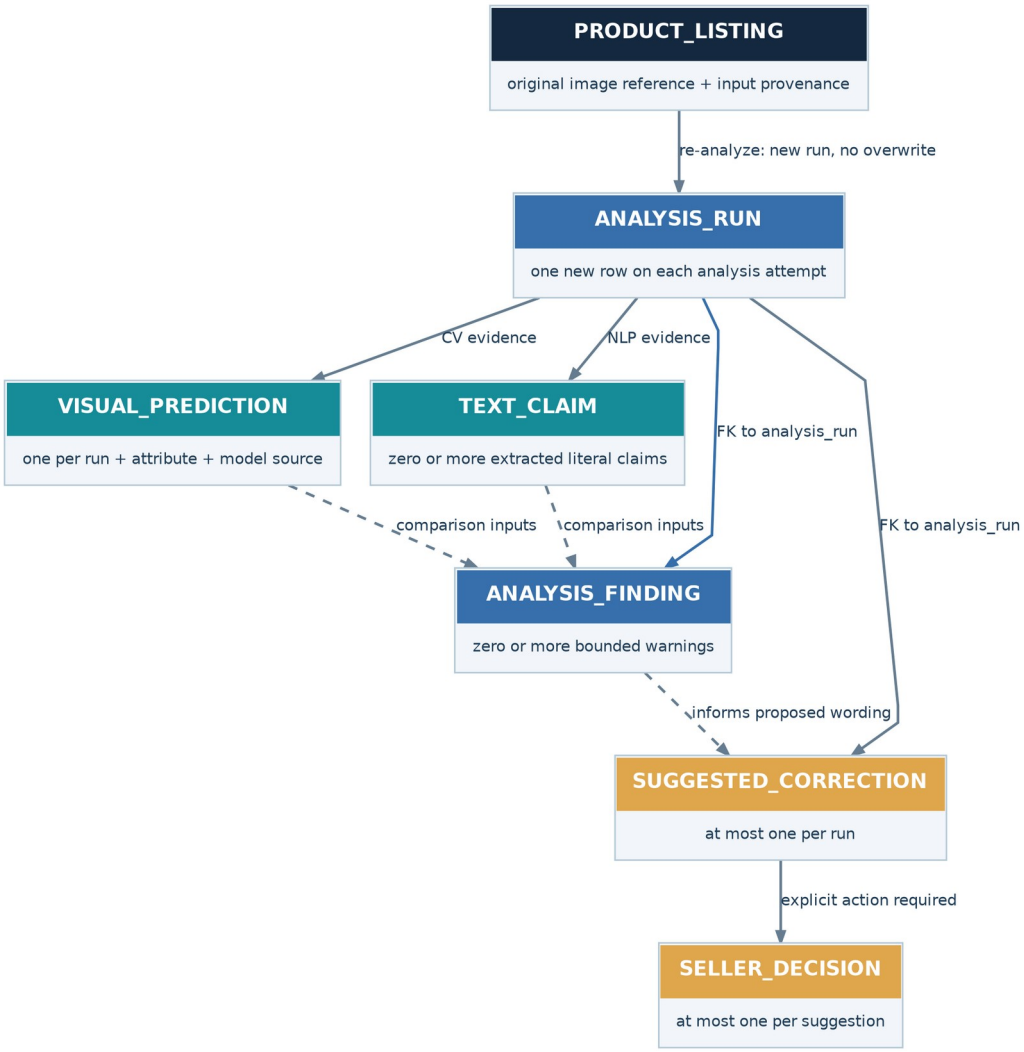


Figure 4. Analysis evidence and repeatability.

ENGINEERING INTERPRETATION

Re-analysis creates another analysis_run instead of overwriting prior records. Predictions and claims are distinct; findings are stored against the run. The visual arrows from predictions/claims to findings represent processing provenance, not SQL foreign keys.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: analysis_run.listing_id; all run-level FKs; one suggested_correction per analysis_run.

Section mapping: 07 Logical relational schema · FR-004 to FR-017; FR-021 to FR-025

DESIGN PLATE / FIGURE 05 OF 08

Human-controlled correction gate

The one-way boundary between suggested wording and an explicit decision.

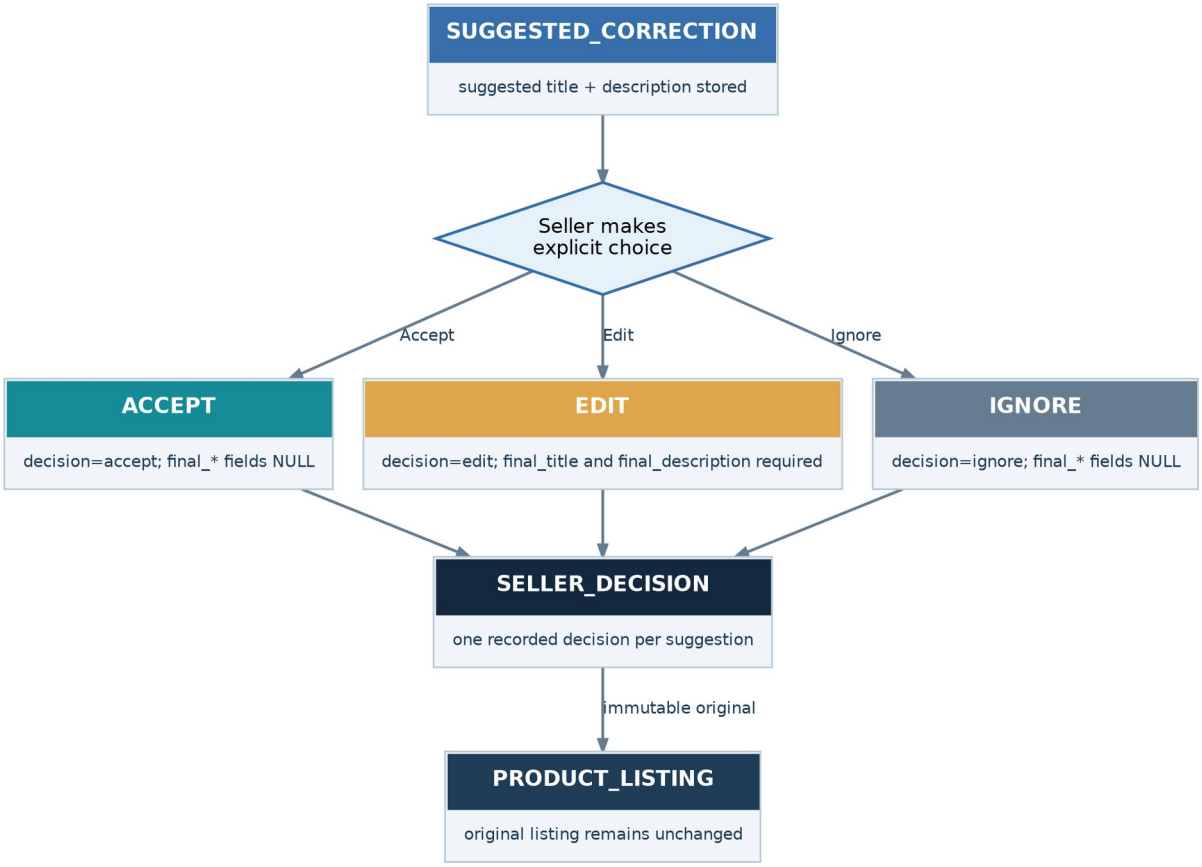


Figure 5. Human-controlled correction gate.

ENGINEERING INTERPRETATION

A suggestion is not an applied update. Accept and Ignore store no final_* values; Edit requires final_title and final_description. This is a business-action map, not an additional state machine table. No publish step exists.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: seller_decision.decision, final_title, final_description; unique(suggestion_id); CHECK ck_seller_decision_02.

Section mapping: 13 Data lifecycle · FR-021 to FR-025

DESIGN PLATE / FIGURE 06 OF 08

Referential integrity and deletion safeguards

Parent-child actions represented by the actual SQL foreign keys.

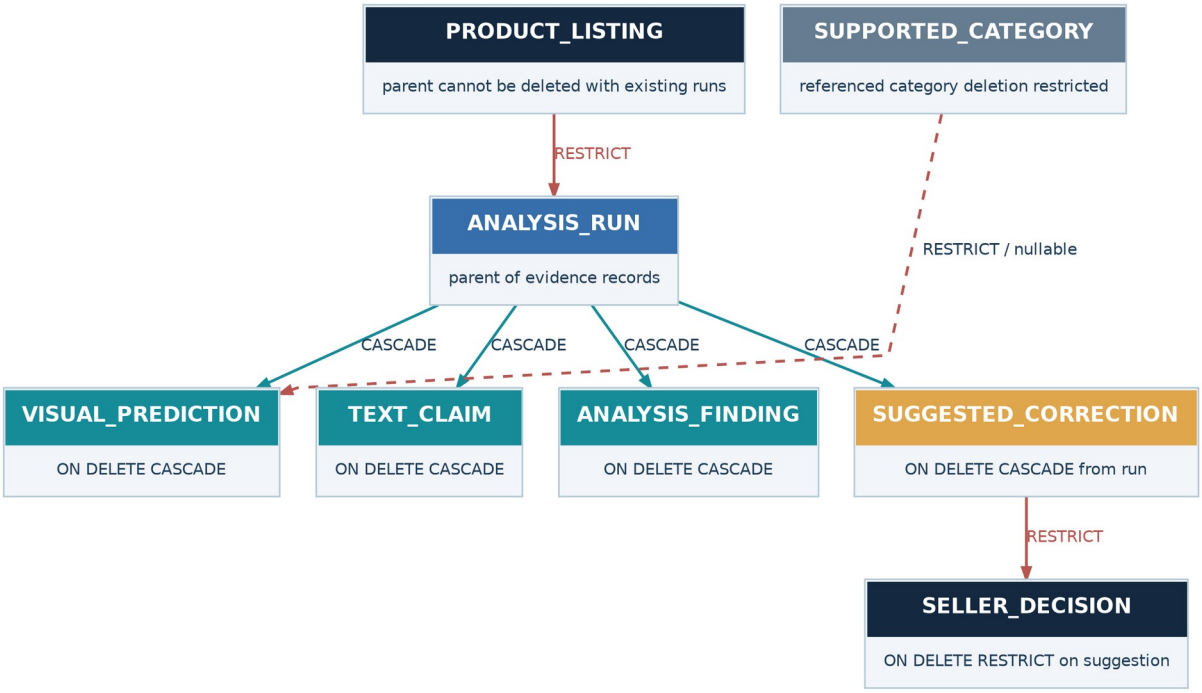


Figure 6. Referential integrity and deletion safeguards.

ENGINEERING INTERPRETATION

Deletion of a product_listing with analysis history is restricted. Removing an analysis_run cascades to evidence children; however, an existing seller_decision restricts deletion of its suggested_correction and therefore can block the upstream cascade. Supported categories referenced by predictions are protected.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: seven foreign keys in SQL/01_create_schema.sql; no soft-delete or purge policy is invented.

Section mapping: 12–13 Data integrity and deletion behavior

DESIGN PLATE / FIGURE 07 OF 08

Normalization pathway to Third Normal Form

From a repeating-group working record to the final eight relations.

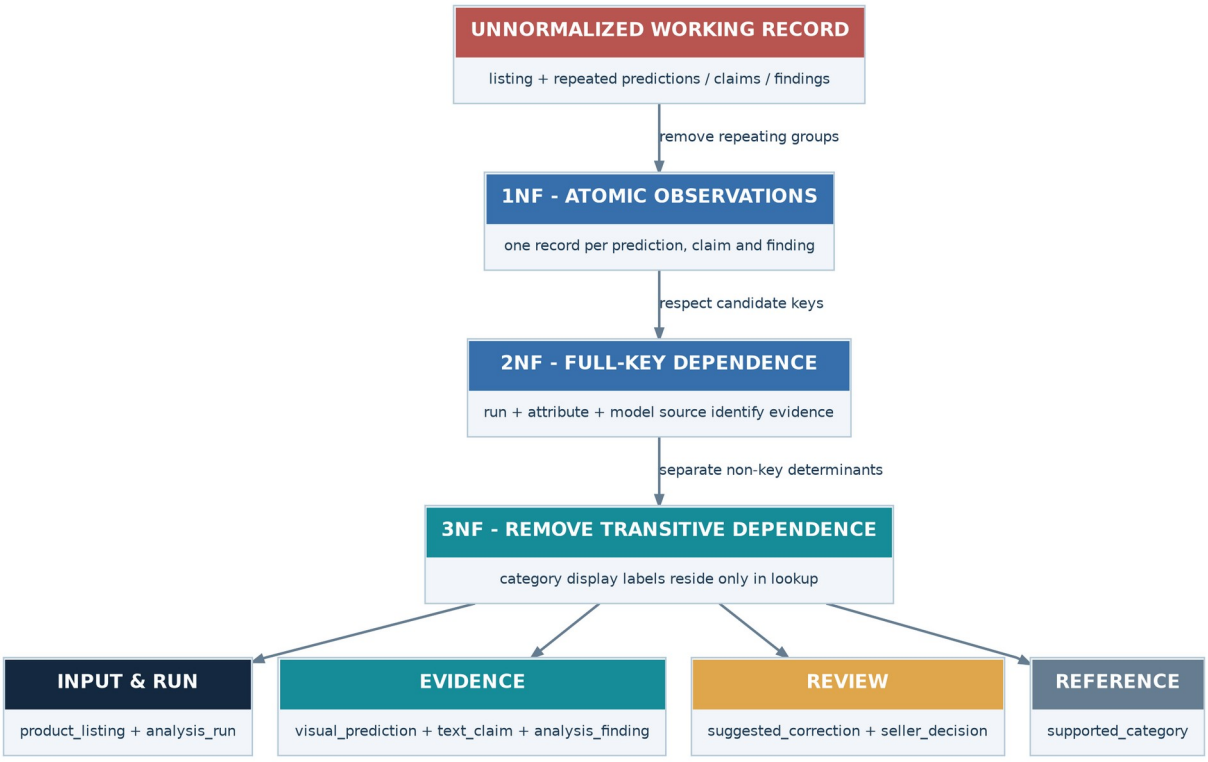


Figure 7. Normalization pathway to Third Normal Form.

ENGINEERING INTERPRETATION

This is a conceptual decomposition argument, not a generated migration. Separating repeated evidence establishes 1NF; enforcing full-key dependence on run, attribute and source supports 2NF; separating category display labels and submitted text avoids avoidable transitive dependencies in 3NF.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: candidate keys and functional dependencies in SmartListing_Normalization_Report.md.

Section mapping: 09 Normalization analysis · 1NF / 2NF / 3NF

DESIGN PLATE / FIGURE 08 OF 08

Derived Review Queue read model

How the reviewer sees the latest relevant status without a ninth table.

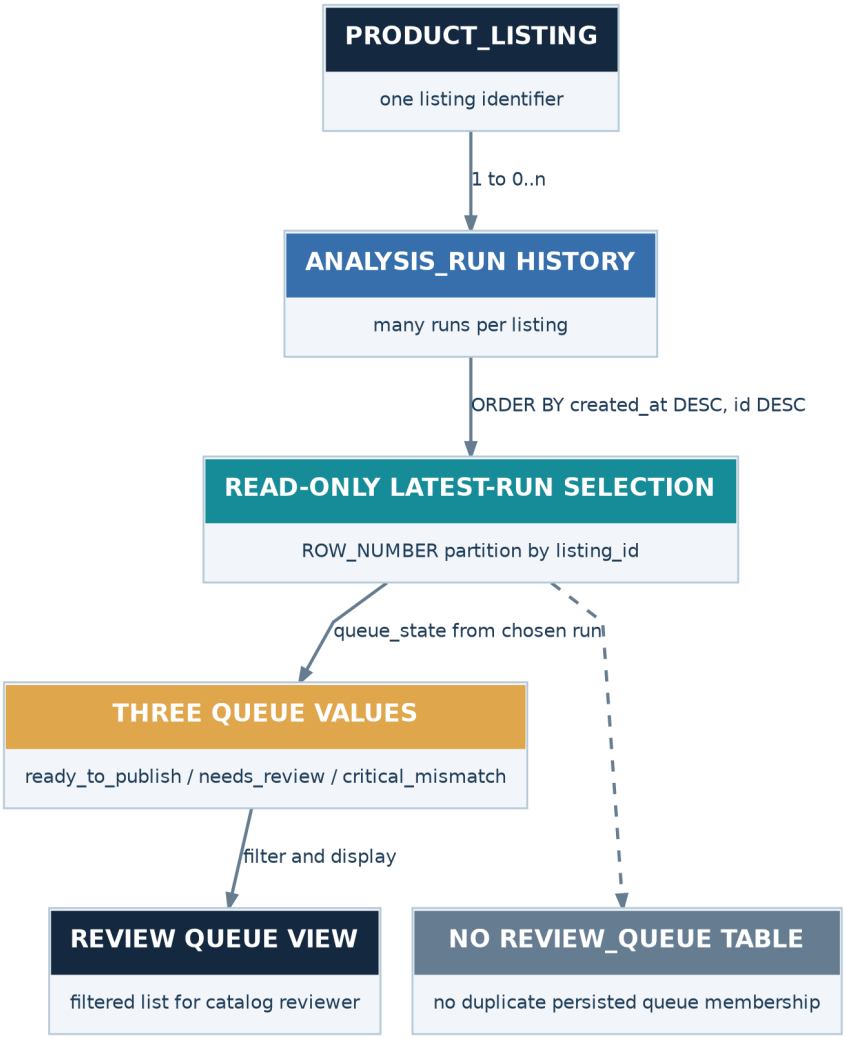


Figure 8. Derived Review Queue read model.

ENGINEERING INTERPRETATION

The queue is a read query against analysis_run. The SQL package contains both a latest-any-run example and an optional completed/partial-only queue example; selecting between them is still an integration decision. Three queue states are stored as values, not separate entities.

SCHEMA / REQUIREMENTS ANCHOR

Evidence anchors: analysis_run.queue_state; SQL/04_validation_queries.sql; FR-018 to FR-020 and UC-13.

Section mapping: 16 Architecture and API alignment · latest-result query